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Comprehensive Analysis of Arabic Sentiment Analysis using Lexicon and Machine Learning based Approaches

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***Abstract*—Arabic sentiment analysis (ASA) is a vital field in Natural Language Processing (NLP) that utilizes Machine Learning (ML) techniques to understand the complex emotional tones that are present in Arabic text. This survey discovers the use of ML in Arabic sentiment analysis, which aims to implement various methodologies for sentiment classification. The goal is to develop a robust model capable of translating nuanced sentiments expressed in different dialects and intricate linguistic structures. The methodology involves creating a labeled dataset, using preprocessing techniques, and utilizing machine learning algorithms such as linear classifiers and deep learning models. By extracting features, training the model, and evaluating its performance, the methodology improves the accuracy of sentiment evaluations. The results of this study highlight the effectiveness of machine learning models in accurately categorizing sentiments in Arabic text, as demonstrated by performance metrics such as precision, recall, and accuracy.**

Keywords—arabic sentiment analysis, natural language processing, linguistic structures, labeled dataset, linear classifiers.

I. Introduction

In the modern world, the internet has transformed into a dynamic platform that allows individuals to freely express their thoughts and share diverse experiences, especially when it comes to products and services. User-generated content has become a valuable resource for decision-making in both public services and business, providing a range of balanced opinions based on personal experiences. Companies recognize the importance of user comments and actively use them to assess customer satisfaction and improve the quality of their offerings. However, sentiment analysis becomes more complex when dealing with the nuanced tones found in Arabic text, as there are limited resources available for this task. Therefore, it is crucial to explore different methodologies to enhance the accuracy of sentiment analysis in Arabic.

This survey aims to tackle the challenging task of sentiment analysis in Arabic text, focusing on the various dialects and intricate linguistic structures that characterize the language. The scarcity of resources for understanding nuanced tones in Arabic text has created difficulties in accurately determining sentiments. To address this gap, the proposed methodology involves Machine learning (ML) techniques in Arabic sentiment analysis. This approach revolves around creating a labeled dataset, applying preprocessing techniques, and utilizing ML algorithms like linear classifiers and deep learning models. The ultimate objective is to develop a robust model that can accurately identify subtle sentiments expressed across different dialects, thereby improving sentiment predictions in Arabic language contexts.

II. Taxonomy of Sentiment Analysis

In this section, the Arabic sentiment analysis is described in the form of a taxonomy that includes algorithms, approaches, and techniques of lexicon-based approach and Machine-learning approach.

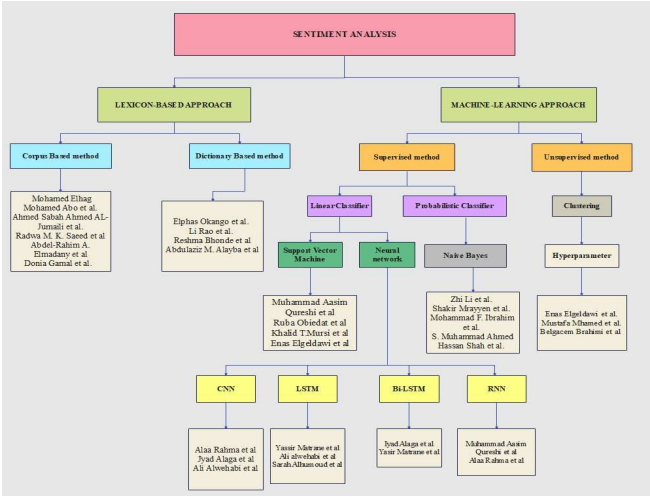

Fig. 1. Taxonomy of SA

Numerous studies have explored sentiment analysis (SA) techniques, often focusing on English and occasionally including other languages such as Arabic, Turkish, Spanish, and Greek. These studies have introduced diverse methods for sentiment analysis for Arabic. In contrast, there is limited research applying sentiment analysis to the Arabic language. The subsequent Section 3 provides details on the dataset,

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including its description and how it is categorized into different classes.

III. Dataset description

The data used for the model were adopted from social media comments, reviews, etc. The dataset contains three sentiment classes: positive, negative, and neutral, providing a complete representation of emotions expressed in Arabic text. It is accurately annotated and balanced, making it a valuable resource for training and evaluating sentiment analysis models.The below table 1 represents the dataset description.

TABLE I. Dataset Description

Dataset	Positive	Negative	Neutral	Total
Twitter [1]	75,774	75,774	-	151,548
Twitter [3]	799	1684	832	3315
Twitter [5]	6151	6146	8767	21,064
Twitter [6]	30,594	30,594	30,587	91,784
Twitter [8]	5497	4503	-	10,000
Twitter [9]	206	605	-	811
Twitter [10]	155	310	35	500
Arabic titles [11]	-	-	-	7500
ArSarcasm [12]	1678	3529	5340	10,547
Twitter [14]	8674	8533	36,082	53,289
ASTD [15]	1642	777	-	2,419
ASTC, ArTwitter, AJGT [16]	30,226	29,420	-	60,546
SAA-SCR [17]	2620	320	800	3740
Hotel review [18]	3500	3500	-	7000
ATDF [19]	93,144	63,263	-	156,407
AMRD [20]	250	250	-	500

IV. Lexicon based approach

Natural language processing can utilize a lexicon-based approach, which involves using predefined dictionaries or lexicons to analyze and understand text. These lexicons consist of word or phrase lists with additional information like meanings or sentiment scores. This approach is commonly used for tasks like sentiment analysis and opinion mining by matching words in the text to entries in the lexicon. The success of this approach depends on the lexicon's quality and coverage. However, challenges arise when dealing with context-dependent meanings and idiomatic expressions, which can be overcome by integrating machine learning techniques for a more detailed language analysis.

A. Corpus method

This method is used in linguistics and NLP to analyze large collections of data called corpora. These corpora serve as datasets for studying language patterns and informing computational models. The main goal of corpus-based is to gain a deeper understanding of language usage, semantics, and structures [1]. The use of this approach is to uncover linguistic characteristics and address challenges in fields such as NLP, and linguistics research. Supervised learning utilizes labeled datasets to train algorithms, enabling accurate data classification and outcome prediction [2]. Arabic sentiment analysis that combines linguistic and statistical features. Linguistics includes stemming and POS tagging, while statistics include TF-IDF [3]. This emphasized the importance of relying on observed data to ensure the integrity of linguistic analyses [4]. Corpora also captures variations in language, enhancing the understanding of linguistic diversity. In natural language processing, corpora are critical resources for training and evaluating models in various tasks. However, challenges such as bias, size and representation, and annotation quality need to be addressed [5]. Overall, corpus-based analysis is a fundamental tool in linguistic research and NLP, providing data for understanding language.

Mohamed Elhag Mohamed Abo et al. (2019) conducted a study that primarily concentrated on examining the different characteristics in the context of Arabic sentiment analysis, and explored different ML methods and algorithms like SVM, NB, DT, k-NN, and supervised approaches. Selecting different samples for corpora ensures capturing real-life language The study intended to address the demand for ASA, especially on media platforms. Also investigated the characteristics, methodologies, and state of sentiment analysis in Arabic, including NLP. The study covered both modern standards and Arabic dialects, using different machine-learning processes. However, handling the diversity of slang proved challenging due to its linguistic complexity.

Ahmed Sabah Ahmed AL-Jumaili et al. (2020) projected a process for ASA that combines linguistic and statistical features with classification methods. The architecture consists of three main components: data, feature extraction, and classification (SVM, KNN, ME). The data focuses on the Arabic corpus used in the research, which includes sentiments collected from Arabic tweets on Twitter. The second component involves feature extraction, using linguistic and statistical. Linguistics includes stemming and POS tagging, whereas statistics include TF-IDF. The experiments used a benchmark dataset of Arabic tweets. Three classifiers, SVM, KNN, and ME, were used. The results exhibited that SVM attained accuracy and f-score due to its ability to handle a large number of classes and its vectorization mechanism, which provides a better representation of the text data.

Radwa M. K. Saeed et al. (2020) present an approach to classify Arabic sentiment using a supervised learning approach. This approach utilizes labeled datasets to train algorithms, enabling accurate data classification and outcome prediction. The study explores two distinct linear transformation methods, namely principal component analysis (PCA) and latent Dirichlet allocation (LDA). The aim is to evaluate the impact of various components on the sentiment classification of Arabic opinion texts. These

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components include negation handling, N-gram feature extraction, feature selection, and positive/negative N-gram feature extraction. Feature selection is performed using PCA and LDA on the negation-handled N-gram features. The proposed method is tested on diverse sentiment problems and datasets, demonstrating improved performance. The results highlight increased accuracy, reduced space usage, and faster classification execution times.

Abdel-Rahim A. Elmadany et al. (2018) presented ArSAS, an Arabic tweet corpus that has been annotated for speech-act recognition and SA. The corpus contains a significant collection of 21k Arabic tweets covering various subjects. It includes annotations for six different speech-act labels. Additionally, sentiment analysis annotations have been added for the same set of tweets, consisting of four classes. The main purpose of this corpus is to encourage research in the fields of speech-act recognition and sentiment analysis specifically for the Arabic language. However, it is important to note that the corpus's applicability to broader contexts may be limited due to the inclusion of multiple Arabic dialects and its focus on specific topics. This could potentially introduce bias, particularly when analyzing sentiment and speech acts in diverse linguistic and thematic landscapes. Therefore, the results obtained from analyzing the corpus may not easily apply or represent sentiments and speech acts in different language contexts or unrelated topics.

Donia Gamal et al. (2019) implemented the use of ML in Arabic Sentiment Analysis by utilizing N-Gram features, which is a collection of elements in a written document that can consist of words, numbers, symbols, and punctuation marks. Over 151,500 tweets/comments in different Arabic dialects were collected and annotated to create a dataset. The development of extracting and classifying entities' emotions and feedback expressed in source text content. Sentiment Analysis (SA) in Arabic faces challenges due to its complex morphology and lack of necessary tools. The aim is to gain a deeper understanding of language usage, semantics, and structures. It compares various SA algorithms, with a particular emphasis on using N-Gram as a feature extraction method. Evaluation metrics like recall, precision, f-measure, and accuracy are used to assess algorithm performance.

This review shows the significance of corpus-based methods in Arabic sentiment analysis is highlighted. The studies analyzed show the effectiveness of using large datasets to improve sentiment classification accuracy. The linguistic nuances in Arabic sentiment and the need for robust corpus-driven approaches are explained. Additionally, the reviewed literature emphasizes the crucial role of corpus-based techniques in achieving nuanced and contextually relevant results in Arabic sentiment analysis.

B. Dictionary-Based Approach

SA is a crucial part of Natural Language Processing (NLP) that uses different methods to understand the emotions expressed in text. Social media provides a platform for analyzing posts, reviews, and feedback to achieve this goal. Sentiment analysis using a dictionary-based approach depends on a pre-existing sentiment lexicon or dictionary. By combining sentiment dictionary information with machine learning and sentiment rules, this method effectively classifies sentiment in review texts and extracts user evaluations on products and topics [6]. The sentiment lexicon contains words or phrases with associated sentiment scores, enabling us to determine whether the text is positive, negative, or neutral [7]. The method involves matching words in the text with entries in the sentiment lexicon and combining the scores to determine the overall sentiment [8]. Initial experiments have utilized various ML algorithms like Naïve Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), deep neural network (DNN), and convolutional neural networks (CNN), to analyze sentiment datasets [9]. While this approach is simple and quick to implement, it does face challenges such as the quality of the sentiment lexicon and sensitivity to context. Nonetheless, sentiment analysis using a dictionary-based approach remains valuable for automatically assessing sentiment in large textual datasets and providing insights for different applications.

Elphas Okango et al. (2022) explored the significance of SA, mainly on Twitter, in understanding the impacts of the COVID-19 pandemic. Analyzing posts, reviews, and feedback on social media provides a valuable opportunity to accomplish this goal. The rise of social media has given people a platform to freely share their thoughts opinions, emotions, and evaluations. The research highlights the potential benefits for governments, such as addressing mental health concerns and supply shortages, while also assessing the effectiveness of guidelines. The importance of sentiment analysis in monitoring public reactions, identifying emerging issues, and detecting misinformation. The objective is to establish a framework for analyzing present social media data, providing insights into the factors that influence the general public during the pandemic, with a specific focus on mental health concerns and limited resources.

Li Rao et al. (2022) have introduced a text sentiment analysis method using a dictionary-based approach that incorporates various features, such as sentiment values from a dictionary, expression features, and an enhanced semantic feature. This approach is specifically designed for colloquial and diverse English social media texts. The proposed method combines sentiment dictionary information with machine learning and sentiment rules to effectively classify sentiment in review texts, to extract user evaluations on products and topics. This study also addresses the limitations of sentiment dictionaries and highlights the importance of integrating multiple features for accurate sentiment analysis. By using a multilevel feature representation method, the researchers have achieved F1 measure and classification accuracy across multiple models

Reshma Bhonde et al. (2015) have explored sentiment analysis using a Dictionary-based approach. This approach involves using dictionaries that contain synonyms and antonyms to compile sentiment words. By utilizing seed sentiment words and bootstrapping from the structure of the dictionary, the analysis aims to determine whether expressed opinions about entities are positive or negative. Sentiment analysis can also be phrase-based, where phrases in sentences are classified based on their polarity. The study highlights how sentiment analysis helps users access relevant information about products and services, enabling them to make informed decisions. Furthermore, the sentiment of an entire document can be summarized as positive, negative, or objective by using a Support Vector Machine (SVM) classifier trained on interpreted sentences for positive and negative classification.

Abdulaziz M. Alayba et al. (2017) have introduced an Arabic dataset that is derived from Twitter, with a specific focus on sentiments related to health services. The paper exactly specifics the data collection process, covering aspects

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of filtering, pre-processing, and annotation. It explores the challenges associated with unstructured Twitter data and underscores the crucial role of effective data preprocessing in Arabic sentiment analysis. The research conducts initial experiments utilizing various machine learning algorithms, including Naïve Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), deep neural network (DNN), and convolutional neural networks (CNN), to analyze sentiments within the health dataset. The insights gleaned from this study contribute significantly to understanding the challenges and methodologies of sentiment analysis, particularly within the realm of Arabic-language data on health services.

The dictionary-based review plays an important role in capturing complex emotional expressions. These lexicons not only offer sentiment polarity but also explore intensity, emotion detection, and contextual variations by using NB, SVM, LR, and CNN.

V. Machine Learning Approach (MLA)

ML approach is transforming the way to understand and engage with different languages, particularly Arabic. By employing machine learning techniques for sentiment analysis in Arabic, we can gain a deeper understanding of human interactions within the linguistic context. This involves creating intelligent models that can accurately extract and analyze emotions and sentiments from texts, leading to advancements in Arabic language processing technology. These advancements have the potential to greatly enhance the automated understanding of Arabic and facilitate effective and intelligent interactions in various applications, including business and social media.

A. Probabilistic Classifier

This classifier is a Machine-Learning (ML) model used in sentiment analysis. It is a part of ML and Natural Language Processing (NLP) and consists of extracting sentiments, emotions, or opinions from written texts [10]. This approach is particularly useful in scenarios like social media monitoring or customer feedback analysis, where it is important to assess the reliability of sentiment predictions [11]. Assigning probabilities to different sentiment classes, such as positive, negative, and neutral, based on patterns it learns from textual data during training. This classifier predicts sentiment labels but also provides confidence scores for each class, giving a more nuanced understanding of its predictions. To conduct the classification process, they evaluate the titles of ideas and dissertations gathered from various sources using standard scraping techniques [12]. The study highlights the importance of understanding characteristics and sentiment expressions in data, laying the groundwork for exploring the intricate relationship between reviews and video content [13].

Zhi Li et al. (2020) have explored into sentiment analysis of danmaku videos using the SD-NB method, which combines a sentiment dictionary and Naive Bayes. This method effectively examines the overall emotional orientation of videos, offering insights into sentiment dimensions and sequential dynamics. By constructing a dedicated danmaku sentiment dictionary, sentiment analysis accuracy is improved, overcoming limitations associated with fixed-length segmentation. Comparative experiments demonstrate the superior performance of SD-NB. The study highlights the importance of understanding characteristics and sentiment expressions in danmaku data, laying the groundwork for exploring the intricate relationship between reviews and video content. The implications extend to effective audience group selection, adjustments in video production, and the advancement of personalized video retrieval approaches. The real-time relevance of danmaku reviews shows promise in enhancing user engagement and enabling personalized recommendations and intelligent services in the network video domain.

Shakir Mrayyen et al. (2018) have focused on Arabic Sentiment Analysis and Opinion Mining. by combining Naive Bayes (NB) with Multilayered Perceptron (MLP) for sentiment analysis, which is a part of machine learning and natural language processing (NLP), consists of extracting sentiments, emotions, or opinions from written content. The use of a Twitter dataset consisting of 2000 labeled tweets, covering various topics like politics and arts, with an equal distribution of positive and negative tweets. They employed different classifiers, including MLP, Decision Table, JRip, OneR, PART, and ZeroR, to classify Arabic sentiment. The study also explored different methods for organizing data into training and testing sets. Overall, the combined MLP classifier demonstrated superior performance in terms of accuracy percentage, making this study a valuable involvement to the field of Arabic sentiment analysis.

Mohammad F. Ibrahim et al. (2021) presented the challenges faced when classifying short texts in the Arabic language. Due to its unique orthographic nature, Arabic requires a specialized preprocessing approach to ensure word meanings are not compromised. The researchers specifically focus on text classification and utilize three standard Naive Bayes classifiers. To improve the classification process, different preprocessing methods are used, including removing punctuation, eliminating stop words, and converting text into space vectors. For feature extraction, the study adopts the TF-IDF method. The implementation of three types of Naive Bayes classifiers revealed that the Naive Bayes Classifier outperforms the others in terms of accuracy during the testing phase. This approach can be applied in diverse scenarios involving the classification of short texts, providing valuable insights into effectively processing the Arabic language for efficient classification.

S. Muhammad Ahmed Hassan Shah et al. (2023) proposed an innovative approach for Arabic Sentiment Analysis and Sarcasm Detection in text classification. This task of assigning categories to text is crucial in natural language processing (NLP), especially for social media monitoring and customer feedback analysis. Traditional NLP struggles with sarcasm, irony, humor, or criticism due to their nuanced nature. To overcome this, researchers introduced a Modified Switch Transformer (MST) for detecting sarcasm and classifying sentiment and dialect in Arabic text data. The MST model incorporates Variational Enmesh Expert's Routing and Probabilistic Projections to enhance accuracy. Tested on the ArSarcasm Dataset, the results indicate the switch transformer excels in detecting sarcasm and performs well in sentiment and dialect classification.

B. Linear classifier

In ASA, the application of linear classifiers within a supervised learning framework emerges as an important approach. In today's social media world, understanding public opinion is crucial for informed decision-making in public services and business [14]. Supervised learning, leveraging labeled data sets, allows linear classifiers to effectively

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differentiate sentiment directions. In the complex context of the Arabic language, characterized by complex morphology and diverse dialects, the role of linear classifiers becomes necessary [15]. This introduction explores the importance and applications of linear classifiers in the field of Arabic sentiment analysis. By combining CNN with hybrid deep learning models, an ensemble model can effectively forecast Arabic sentiment with precision [16]. Utilizing the Support Vector Machine (SVM) can help to review texts as positive, negative, or neutral [17]. Highlighting the ability to provide accurate and understandable sentiment predictions. In the exploration of complexities of sentiment analysis in Arabic text, linear classifiers serve as fundamental tools, contributing to a nuanced understanding of the emotions expressed in the linguistic domain.

Hager Saleh et al. (2022) have projected a new way to tackle the challenges of sentiment analysis in Arabic content on social media platforms. The acknowledgment of the significance of analyzing people's opinions from these platforms and focus on enhancing Arabic Sentiment Analysis (SA). Due to the intricate morphological structures and diverse dialects of the Arabic language, sentiment analysis remains difficult. To address this, the proposed model utilizes machine learning and deep neural algorithms, introducing a Heterogeneous Ensemble Deep Learning approach. This ensemble combines Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) architectures to enhance the accuracy of sentiment analysis predictions. The study demonstrates that the ensemble model outperforms other approaches across various datasets, highlighting its effectiveness in capturing the intricacies of sentiment in Arabic text on social media. This research lays the groundwork for further advancements in sentiment analysis methodologies tailored to the specific nuances of the Arabic language on digital platforms.

Ali Louati et al. (2023) have proposed the realm of sentiment analysis with a focus on Arabic course reviews in the context of a Saudi university. The study employs machine learning, specifically utilizing the Support Vector Machine (SVM) algorithm, to classify reviews into three classes. The classifier is trained on Arabic text, providing a system to assess sentiment. Metrics such as accuracy, precision, and recall are used to evaluate the effectiveness of the SVM-based model. Valuable insights are derived from this research, shedding light on the difficulties that Arab universities encounter after the COVID-19 pandemic. The classified student reviews serve as a tool to identify areas for improvement in the educational process. In addition, the system helps identify specific challenges that students face, thereby providing targeted solutions to address these issues.

Arief Setyanto et al. (2022) contribute to the domain of opinion mining with a focus on the Arabic language using Long Short-Term Memory (LSTM). In the contemporary landscape of textual communication dominated by social media, understanding public opinion is crucial for making informed decisions in various fields, including public services and business. The study explores various transformation algorithms, ranging from traditional methods TF-IDF to advanced word embedding techniques such as GloVe, word2vec, BERT, and fastText. The experimentation involves applying these algorithms to transform an Arabic text dataset, and the research compares their accuracy for opinion mining using the ASAD dataset which is annotated tweets across three classes. The evaluation results show the effectiveness of the triple-layer LSTM model with fastText word embedding, by its superior testing accuracy. This research arranges the basis for enhancing opinion mining methodologies in the context of the Arabic language, offering insights into effective approaches for sentiment analysis and opinion polarity assessment.

Hager Saleh et al. (2022) have proposed a significant contribution to Arabic Sentiment Analysis (SA) by introducing an Enhanced Sentiment Analysis method. This method utilizes a unique Stacking Ensemble of Hybrid and Deep Learning Models. SA is a crucial application in machine learning and NLP that focuses on extracting opinions from text. Arabic SA encounters several challenges, including dialectical diversity, morphological complexities, and the implicit nature of sentiment terms. Hybrid models, specifically those that combine CNN with LSTM or GRU, have demonstrated improved performance. In their study, the authors propose a stacking ensemble model that combines CNN with hybrid deep learning models to accurately predict Arabic sentiment. The approach involves optimizing three deep learning models in the first phase and integrating their outputs using a Support Vector Machine (SVM) in the second phase. The proposed deep stacking model exhibits superior performance, providing a robust and accurate Arabic sentiment analysis.

C. Unsupervised Learning

Unsupervised learning is an influential pattern in machine learning, which involves exploring patterns and structure within data without any help in the form of labeled datasets. Unsupervised learning is particularly effective in clustering, where similar data are grouped on similarities. In the background of Arabic sentiment analysis, utilizing an unsupervised learning approach becomes important especially when dealing with large datasets in a complex language like Arabic [18]. The introduction of hyperparameter tuning further enhances the performance of unsupervised learning models in sentiment analysis, as optimal configurations ensure more accurate representations of sentiment nuances in the Arabic language. Hyperparameters can be implemented by using the TensorFlow framework [19]. Extracting opinions is important to improve opinion classification in reviews by using the n-gram word approach and the POS tagging approach [20]. The combination of unsupervised learning for clustering and fine-tuned hyperparameters exemplifies a robust approach to unveiling hidden structures and sentiments within Arabic text data.

Enas Elgeldawi et al. (2021) presented the field of Arabic Sentiment Analysis (SA) by proposing a method to expand the accuracy of machine learning algorithms through hyperparameter tuning. This is particularly important for businesses that depend on customer comments and reviews. Analyzing Arabic text, which has complex morphology, presents multiple challenges, and machine learning plays a critical part in addressing them. The paper conducts a comprehensive analysis of different hyperparameter tuning techniques. The focus is on optimizing the precision of well-known machine learning classifiers such as Logistic Regression (LR), Ridge Classifier (RC), Support Vector Machine Classifier (SVC), Decision Tree (DT), Random Forest (RF), and Naive Bayes (NB). These classifiers are applied to an Arabic sentiment classification problem, which tackles the difficulty of extracting sentiment from the intricate

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Arabic language. The study estimates the presentation of each hyperparameter tuning technique using a constructed dataset, providing valuable insights into their strengths and limitations in enhancing classifier performance.

Mustafa Mhamed et al. (2021) presented the advancement of Arabic sentiment analysis by introducing improvements through a CNN-based architecture and text preprocessing techniques. Recognizing the extensive expression of personal emotions and attitudes on social media platforms like Facebook, Twitter, and Instagram, the work emphasizes the importance for governments to analyze public opinion. The study specifically applies two models to ASTD and ATDFS datasets, employing text cleaning and preparation methods tailored to the challenges posed by Arabic tweets. Tokenization is performed using the Natural Language Toolkit (NLTK), with subsequent application of methods MC1 and MC2 for multiclass and binary classification. The research involves tuning hyperparameter values and utilizes the TensorFlow framework for implementation. By addressing the nuances of Arabic sentiment analysis, this work contributes to a better understanding of public opinion in the digital world.

Belgacem Brahimi et al. (2019) have presented Sentiment analysis in Arabic by introducing improvements using a combined approach. This approach is mostly used on social networking spots and media platforms. The study proposed methods for extracting opinions to improve opinion classification in reviews by using the n-gram word approach and the Parts-of-Speech tagging approach. These two methods are combined in this paper to enhance performance on classification. To improve the performance of the feature reduction technique by using a (SVM) which is an algorithm in sentiment classification. The proposed combined approach provides an effective and better score than baseline sentiment classifiers.

VI. Comparison Table

Table 2. Shows Advantages and Limitations of studies in Arabic sentiment Analysis

TABLE II. Advantages and Limitations of Studies

Author	Advantage	Limitation
Donia Gamal et al. [1]	It examines different algorithms using n-gram as a feature extraction technique and finds that unigram combined with either Passive Aggressive (PA) or Ridge Regression (RR) achieves the highest accuracy.	The findings may not apply to other contexts because they only focus on Twitter posts in Arabic dialects. The effectiveness of the algorithms may also differ on different social media platforms or types of text.
Radwa M. K. Saeed et .al [2]	The optimized features, negation handling, and effective feature reduction techniques lead to significant improvements in accuracy compared to existing works.	Limited exploration of its applicability to other languages, along with a limited range of evaluation metrics, may hinder the comprehensive assessment and broader usability of the approach.
Ahmed Sabah Ahmed AL-Jumaili et al. [3]	SVM attained accuracy and f-score due to its ability to handle a large number of classes and its vectorization mechanism, which provides a better representation of the text data.	The paper fails to address any potential biases in the benchmark dataset and does not discuss the generalizability of the linguistic and statistical features to domains other than Arabic tweets. This limits the wider application of the proposed hybrid method.
Mohamed Elhag Mohamed Abo et al [4]	This examines the latest machine learning-based sentiment analysis techniques for Arabic dialects, providing a detailed overview of methods and classifiers such as SVM, NB, DT, and k-NN. It also discusses trends and challenges in the field.	MSA in sentiment analysis studies may restrict the generalizability of the findings to different Arabic dialects. Moreover, the complexity of handling diverse Arabic slang presents a challenge, which could impact the coverage and accuracy of sentiment analysis for informal expressions.
AbdelRahim A. Elmadany et al. [5]	The dataset's inclusion of various dialects, avoidance of sentiment keywords, and high inter-annotator agreement make it an excellent resource for advancing Arabic sentiment analysis and speech act recognition research.	The use of annotation introduces bias in labeling, which may affect the dataset's reliability. Furthermore, the corpus's focus on controversial topics might not fully capture the range of sentiments.
Elphas Okango et al. [6]	It helps governments tackle mental health concerns and supply lacks by analyzing expressions from Twitter users.	The findings may not accurately represent the entire population due to biases in social media user demographics and limited consideration of regional differences
Reshma Bhonde et al. [7]	Providing consumers with a valuable tool to make informed choices and gain quick insights through graphical analysis.	When graphical representations are lacking or insufficient, and the accuracy of automated text analysis techniques may be limited by the complexity of opinions expressed, which can hinder nuanced understanding.
Li Rao et.al [8]	By combining sentiment value, expression, and improved semantic features. This integration results in improved metrices.	However, the method's reliance on diverse features may lead to increased computational complexity. Additionally, the paper lacks a thorough discussion on potential challenges or biases in sentiment analysis, which limits the overall understanding of the method's applicability.
Abdulaziz M. Alayba et al. [9]	By utilizing a health services dataset, implementing different machine learning algorithms, and obtaining impressive accuracy outcomes	limitations to consider, such as possible bias in manual annotation and the absence of complete performance metrics beyond accuracy. These limitations may affect the reliability of the dataset

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		and the overall evaluation of sentiment analysis techniques.
Shakir Mrayyen et al. [10]	The hybrid Naive Bayes-Multilayered Perceptron approach provides a practical solution for effective opinion classification in decision-making contexts.	Insufficient insight into MLP components, and no discussion of computational complexity or misclassification scenarios.
S. Muhammad Hassan Shah et al. [11]	By using the switch transformer, it exceeds other models in sarcasm detection and demonstrates strong performance in sentiment classification and dialect categorization.	Lacks evidence of the model's generalization to diverse datasets and computational efficiency details, and the comparative analysis could be broadened to include a more comprehensive set of state-of-the-art models.
Mohammed F. Ibrahim et al. [12]	This approach is evident in its ability to achieve accurate text classification in Arabic, especially when utilizing the CNB classifier. Additionally, it shows potential in document classification based on titles, providing practical solutions for short text classification.	There are several limitations to consider when dealing with Arabic text, such as difficulties in managing spelling and language differences. Additionally, the model may be influenced by incomplete or biased training data, and its effectiveness in different domains and languages may be uncertain.
Zhi Li et al. [13]	By creating a sentiment dictionary and using Naive Bayes. It shows success in extracting emotional information, sentiment classification, and popularity prediction in danmaku videos.	The method's usefulness may be restricted to danmaku data and may not easily apply to other types of text data..
Arief Setyanto et al. [14]	It provides a thorough inspection of word embedding methods and LSTM architectures, offering insights for effective opinion mining in Arabic text.	Absence of details on dataset augmentation challenges and limited generalizability due to the specific characteristics of the ASAD dataset
Hager Saleh et al. [15]	The ensemble model demonstrates superior performance compared to alternative approaches across diverse datasets, underscoring its efficacy in capturing the complexities of sentiment in Arabic text.	It does not provide insights into the interpretability of the suggested ensemble model, limiting the understanding of how it makes decisions in Arabic sentiment analysis.
Hager Saleh et al. [16]	Combining pre-trained DL models and meta-learners, showcasing enhanced Arabic sentiment analysis performance.	It does not provide insights into the interpretability of the proposed model, limiting its transparency and generalizability.
Ali Louati et al. [17]	SVM-SAA-SCR algorithm, outperforming CAMeLBERT and providing valuable insights for enhancing post-COVID-19 educational experiences in Arab universities.	The level of uncertainty surrounding the ability to navigate cultural nuances in the Arabic language is still unclear.
Enas Elgeldawi et al [18]	The accuracy of the Support Vector Machine Classifier in Arabic sentiment classification can be greatly improved through comprehensive hyperparameter tuning, especially when utilizing Bayesian Optimization.	The effectiveness of tuning approaches, particularly for Naive Bayes, may not apply to different datasets or scenarios.
Mustafa Mhamed et al [19]	By employing two models with innovative preprocessing techniques, outperforming existing benchmarks and emphasizing the importance of text preprocessing for better performance.	The limited scope of specific datasets and the lack of discussions on generalizability might restrict the wider application of the proposed preprocessing methods and models
Belgacem Brahimi et al. [20]	Various techniques such as n-gram and skip-n-gram models, Part-Of-Speech tagging, and feature reduction, to effectively analyze sentiment in Arabic movie reviews. These methods have shown significant enhancements, surpassing baseline classifiers.	It is mainly limited to movie reviews, and the applicability of the proposed methods to different domains is not thoroughly investigated. so it lacks specific information regarding potential obstacles or factors to consider.

VII. Conclusion

This survey addressed the contribution to the field of sentiment classification for Arabic reviews by introducing various methods to extract relevant opinions and improve classification outcomes. The integration of these methods has shown improvements in classification performance. The study focused on the challenges of retrieving Arabic dialects from social media platforms like Facebook and Twitter and placed the literature review within the context of Arabic dialect sentiment analysis. The exploration of state-of-the-art machine learning-based Arabic sentiment analysis highlighted the effectiveness of classifiers such as SVM, NB, DT, k-NN, and supervised approaches, especially for unstructured Arabic language. Comparisons between Modern Standard Arabic (MSA) and Arabic dialects emphasized the subtle differences in sentiment expression. Traditional sentiment analysis methods were separated into dictionary-based and machine learning-based approaches, each with its limitations. The proposed method for text sentiment analysis, which integrates multiple features, offers an approach to build a robust text sentiment classification model.

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